

Technical Document #521: Retrieval Augmented Generation - Comprehensive Overview

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Query decomposition breaks complex queries into simpler sub-queries. It improves retrieval quality for multi-faceted questions.

Chunking splits documents into smaller segments for embedding. Optimal chunk size balances context retention with retrieval precision.

Hallucination detection identifies when LLMs generate factually incorrect information. It is critical for maintaining trust in RAG systems.

Vector databases store dense embeddings for semantic search. Pinecone, Weaviate, and ChromaDB are popular vector database solutions.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

The rapid advancement of technology has transformed how organizations approach problem-solving and innovation. Companies must adapt to stay competitive in an ever-evolving landscape.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Key Takeaways:

- Semantic search finds documents based on meaning rather than exact keywords.
- Re-ranking models score retrieved documents for relevance to the query.
- Context window optimization selects the most relevant information to fit within the LLM's context limit.